

Module 10 - Activity Feed & Notifications

Audit-style collaboration events and user alerts

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Create a lightweight activity feed for groups/plans and a user notification center. Activity is primarily a collaboration history; notifications are user-targeted alerts.

Activity document

```
groups/{groupId}/activities/{activityId}
{
  actorId,
  type,
  message,
  entityType,
  entityId,
  planId,
  createdAt
}
```

Notification document

```
users/{uid}/notifications/{notificationId}
{
  type,
  title,
  body,
  groupId,
  planId,
  entityId,
  readAt,
  createdAt
}
```

Events to support

- Group created
- Member joined
- Member removed
- Plan created
- Plan updated
- Booking added
- Expense added
- Expense settled
- Poll created
- Poll closed
- Invitation used

Implementation rule

Generate audit-style activity from the action service or trusted function, not from arbitrary UI text. Messages should be deterministic and generated from structured event types.

Acceptance criteria

- Activity is newest-first.
- Notifications show unread count.
- Mark-read is optimistic with rollback on error.
- Opening a notification can navigate to the relevant entity.